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Inventor(s)	Burch; Eric

Universal connector for adjustable golf clubs

Abstract

A universal connector for an adjustable golf club is provided. Such a universal connector includes a first member having a head portion and a body portion and having a bore for receiving a terminal end of a shaft. The connector also includes a second member having a bore for receiving a portion of the first member therein. The first member includes a first anti-rotational member and the second member includes a second anti-rotational member. The first and second anti-rotational members are engageable with one another to prevent relative rotation of the first member relative to the second member about a longitudinal axis of the connector. The second member includes a third anti-rotational member, the third anti-rotational member configured to mate with an anti-rotational member of the golf club head to prevent relative rotation of the second member relative to the head.

Inventors:	Burch; Eric (Rockford, IL)
Applicant:	Club-Conex LLC (Loves Park, IL)
Family ID:	1000008821492
Assignee:	Club-Conex LLC (Loves Park, IL)
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Primary Examiner: Passaniti; Sebastiano

Attorney, Agent or Firm: Reinhart Boerner Van Deuren P.C.

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This patent application is a continuation of co-pending U.S. patent application Ser. No. 15/639,303, filed Jun. 30, 2017, which is a continuation of U.S. patent application Ser. No. 14/226,519, filed Mar. 26, 2014, which issued

on Aug. 8, 2017, as U.S. Pat. No. 9,724,571, the entire teachings and disclosure each of which are incorporated herein by reference thereto.

FIELD OF THE INVENTION

(1) This invention generally relates to golf equipment and more specifically to connectors for interchangeable golf club heads and golf club shafts.

BACKGROUND OF THE INVENTION

(2) The typical way to connect a golf club shaft and a golf club head is to epoxy the tip end of the golf club shaft into a bore formed within the hosel portion of the golf club head. Many golf shops offer a broad range of exotic shafts and club heads to meet the standards of today's demanding golfer.

(3) To facilitate the ease in which various combinations of shafts and heads may be evaluated by the golfer, some golf equipment makers, such as Adams Golf, Callaway, Cobra, Cleveland, Mizuno, Nike, Taylor Made, Titleist and many others have devised connectors that rapidly secure a shaft and a club head together thereby making such clubs are referred to as "adjustable" clubs. Because these connectors are releasable, if the customer doesn't like the existing shaft and head combination the connector is simply released and a different shaft and head combination is secured for evaluation by the golfer.

(4) In such adjustable club configurations, the connector used thereby is typically mounted on the club head and contains brand specific mounting features thereon.

(5) In recent years there has been an increase industry wide of such connectors. As a result, each manufacturer that utilizes adjustable connectors has elected to employ a proprietary connector configuration that is not interchangeable with other brands of equipment. Some have "squared keyways", "hexagon keyways", "splines", etc., and as a result. The result of such a configuration limits a user to only interchanging club heads and club shafts that are of the same brand. However, there are many instances where a user may prefer one brand of club shafts that is not the same brand as that user's preferred brand of club heads. In such instances, such users have heretofore been prevented from utilizing such a preferred combination because the brand specific connector portion situated on the club head will not mate with the differing brand specific connector portion situated on the club shaft. Accordingly, there is a need in the art for a universal connector for adjustable golf clubs.

(6) The invention provides such a universal. These and other advantages of the invention, as well as additional inventive features, will be apparent from the description of the invention provided herein.

BRIEF SUMMARY OF THE INVENTION

(7) In one aspect, the invention provides a universal connector for connecting a golf club head having a hosel and a golf club shaft, regardless of the fact that the golf club head and shaft are of the adjustable construction and utilize a brand specific connector arrangement. Such a universal connector includes a first member having a head portion and a body portion and having a bore for receiving a terminal end of a shaft. The connector also includes a second member having a bore for receiving a portion of the first member therein. The second member has a head portion and a body portion. The body portion is sized to be received within a bore of the hosel. The first member includes a first anti-rotational member and the second member includes a second anti-rotational member. The first and second anti-rotational members are engageable with one another to prevent relative rotation of the first member relative to the second member about a longitudinal axis of the connector. The second member includes a third anti-rotational member, the third anti-rotational member configured to mate with an anti-rotational member of the golf club head to prevent relative rotation of the second member relative to the head.

(8) In another aspect, the invention provides a universal connector for connecting a golf club head having a hosel and a golf club shaft, which such a connector providing a reduced part count and

lower cost alternative than current designs. The connector includes a first member having a first anti-rotational member formed thereon and configured for connection to the shaft. The connector also includes a second member having a second anti-rotational member formed thereon and configured for being received within a bore of the hosel, the first and second anti-rotational members engageable with one another. The first member includes a threaded bore configured for receipt of a mounting screw extending through a portion of the club head, wherein tightening of the screw axially biases the first and second members together within the hosel.

(9) In certain embodiments according to the foregoing aspects, the first anti-rotational member is a plurality of downwardly extending teeth formed at an axial extent of the head portion of the first member. The second anti-rotational member is a plurality of upwardly extending teeth formed at an axial extent of the head portion of the second member. The downwardly extending teeth are received within spaces formed between the upwardly extending teeth to rotationally fix the first member relative to the second member.

(10) In certain embodiments according to the foregoing aspects, the head portion of the first member has a maximum outer diameter which is greater than a maximum outer diameter of the body portion of the first member. The body portion of the first member includes a transition region and an extension region. A first axial abutment surface is formed at an interface of the transition region and the extension region. A second axial abutment surface is formed along the bore of the second member and is configured to axially abut the first axial abutment surface.

(11) In certain embodiments according to these aspects, the third anti-rotational member is selected from the group consisting of: a plurality of radially outwardly extending ribs formed at an end of the body portion of the first member, a plurality of downwardly extending teeth formed at an interface of the head portion and the body portion of the first member, a plurality of radially outwardly facing facets formed at the interface between the head portion and the body portion of the first member, a plurality of downwardly extending teeth formed at an axial extent of the head portion of the first member, a plurality of outwardly extending ribs formed along an outer periphery of the body portion of the first member between the head portion and an end of the body portion.

(12) In yet another aspect, the invention provides a method for installing a universal connector on a golf club having a head with a hosel and a shaft. The method includes inserting a portion of the shaft into a bore of a first member of the connector. The method also includes inserting a portion of a second member of the connector into a bore of the hosel. The method also includes inserting a portion of the first member into a bore of the second member such that a portion of the first member and a portion of the shaft received in the bore of the first member are positioned within the bore of the second member. The method also includes rotationally fixing the second member relative to the club head and rotationally fixing the first member relative to the second member.

(13) The method can also include securing the portion of the shaft within the bore of the first member using an adhesive.

(14) In certain embodiments, the method can also include rotationally fixing the first member relative to the second member by bringing a first anti-rotational member of the first member into engagement with a second anti-rotational member of the second member.

(15) In certain embodiments, the method can also include rotationally fixing the second member relative to the club head by bringing a third anti-rotational member of the second member into engagement with a club head anti-rotational member formed on the hosel of the club head.

(16) Other aspects, objectives and advantages of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings incorporated in and forming a part of the specification illustrate several aspects of the present invention and, together with the description, serve to explain the principles of the invention. In the drawings:
- (2) FIG. 1 is a perspective view of one embodiment of a universal connector according to the teachings of the invention shown in an assembly view with a golf club head and a partially shown golf club shaft;
- (3) FIG. 2 is a perspective exploded view of the embodiment shown in FIG. 1, showing first and second members of the connector;
- (4) FIG. 3 is a perspective exploded view of the embodiment shown in FIG. 1 illustrated in cross section;
- (5) FIGS. 4-5 are perspective views of another embodiment of a second member of a connector according to the teachings of the invention;
- (6) FIGS. 6-7 are perspective views of another embodiment of a second member of a connector according to the teachings of the invention;
- (7) FIGS. 8-9 are perspective views of another embodiment of a second member of a connector according to the teachings of the invention; and
- (8) FIGS. 10-11 are perspective views of another embodiment of a second member of a connector according to the teachings of the invention.
- (9) While the invention will be described in connection with certain preferred embodiments, there is no intent to limit it to those embodiments. On the contrary, the intent is to cover all alternatives, modifications and equivalents as included within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

- (10) Turning now to the illustrations, several embodiments of a connector for an adjustable golf club are illustrated. As will be explained in greater detail below, these embodiments overcome existing problems in the art discussed above by providing a universal connector that will readily work with a variety of differing brand specific connectors. Indeed, embodiments of the universal connector described herein allow a user to connect a club shaft of one brand with a club head of another brand, despite the fact that the originally provided connector componentry for the club shaft and the club head do not mate with one another.
- (11) Turning now to FIG. 1, a first embodiment of a connector **20** is illustrated. Connector **20** connects a golf club shaft **22** to a golf club head **24**. As is known in the art, club head **24** includes a hosel **28** extending therefrom which defines a bore **26** extending into club head **24**. A portion of connector **20** is received within bore **26**. Another portion of connector **20** is secured to club shaft **22**. The portion of connector **20** which is received in club head **24** is secured to club head **24** using a mounting fastener **32**. Club head **24** is generally illustrated as an adjustable club head in that it includes brand specific features in the area of hosel **28** for receiving a portion of a brand specific connector. As will be explained in greater detail below, embodiments of the invention described herein provide a connector which includes a portion thereof designed to mate with the aforementioned brand specific features of the connector originally provided with club head **24**. This portion of the connector described herein also provides a universal interface such that a common other portion of connector **20** may be utilized on club shaft **22** so that any club shaft **22** utilizing this other portion of the connector described herein will mate with the portion of the connector received in club head **24**, regardless of the fact that club shaft **22** and club head **24** are of different brands.
- (12) With specific reference now to FIG. 2, connector **20** includes a first member **36** and a second member **38**. First member **36** includes a bore **44** which receives a portion of club shaft **22**. Second member **38** also includes a bore **46** which receives a portion of first member **36**, as well as some

extent of club shaft **22** which is received in first member **36**.

(13) First member **36** may be readily secured to club shaft **22** through the use of adhesives. Additionally, first member **36** may also include alignment indicia thereon for rotationally aligning first member **36** relative to connector axis **30** relative to club shaft **22**.

(14) First member **36** also includes a first anti-rotational member **52**. Second member **38** includes a second anti-rotational member **54**. The first and second anti-rotational members **52**, **54** engage one another in the fully assembled state of connector **20** to prevent rotation of first member **36** relative to second member **38** and vice versa.

(15) Additionally, second member **38** also includes a third anti-rotational member **56** which mates with a club head anti-rotational member **58** disposed on or within the hosel **28**. Third anti-rotational member **56** and club head anti-rotational member **58** engage one another to prevent rotation of second member **38** relative to club head **24** rotationally about connector axis **30**. As will be described in greater detail below, club head anti-rotational member **58** will vary depending upon the specific brand of club head. As such, the particular structural attributes of second member **38** will vary depending upon the particular club head **24**. However, second member **38** will always provide a universal second anti-rotational member **54** and bore **46** to accommodate a single embodiment of a first member **36**. As a result, a user may utilize the same first member with differing second members. As a practical result, a user may utilize a single set of preferred golf club shafts **22** which incorporate first member **36** with a plurality of club heads **24** of differing brands. All that is required to achieve this functionality is the utilization of the brand specific second member **38**, and more particularly its specific configuration of a third anti-rotational member **56** designed to mate with the particular configuration of a club head anti-rotational member **58** of club head **24**.

(16) Indeed, first anti-rotational member **52** of first member **36** is embodied by a plurality of downwardly extending teeth **60** as shown in FIG. 2. Those skilled in the art will readily recognize that the particular number of teeth shown should be taken by way of example only, as the number of teeth **60** may be varied. Second anti-rotational member **54** of second member **38** is defined by a plurality of upwardly extending teeth **62** which mate with downwardly extending teeth **60**. These upwardly extending teeth **62** are common across all embodiments of second member **38**, and thus all of these embodiments readily mate with the illustrated embodiment of first member **36**.

(17) In the particular embodiment illustrated, third anti-rotational member **56** is embodied by a plurality of radially outwardly extending ribs **64**. These ribs **64** mate with radially inwardly extending ribs formed along the interior surface of bore **26** of hosel **28** which form the club head anti-rotational member. The engagement between the aforementioned ribs prevents rotation of second member **38** relative to club head **24** about connector axis **30**.

(18) Turning now to FIG. 3, the structural attributes of first and second members **36**, **38** are described in greater detail. As can be seen in this view, a portion of shaft **22** is received within bore **44**. Shaft **22** extends into bore **44** until it positively engages a bottom of bore **44**, however, other depths of insertion may be utilized. As described above, shaft **22** may be secured within bore **44** uses an adhesive or any other known method. First member **36** includes a head portion **66** and a body portion **68** which extends away from head portion **66**. As illustrated, the head portion **66** has a tapered portion **67** with an outer radial surface that reduces in outer diameter when moving axially away from the first anti-rotational member **52**. The tapered portion **67** being axially offset from at least a portion of the body portion **68**. The tapered portion **67**, body portion **68**, and anti-rotation member **52** are formed as a single piece. Body portion **68** includes a transition region **70** and an extension region **72**. As shown, head portion **66** has a maximum outer diameter which is greater than a maximum outer diameter of body portion **68**. Additionally, transition region **70** has a greater maximum outer diameter than a maximum outer diameter of extension region **72**. A first axial abutment surface **74** is formed at the interface between transition region **70** and extension region **72**. This first axial abutment surface **74** axially abuts a second axial abutment surface **84** formed

within the bore **46** of second member **38** when connector **20** is completely assembled.

(19) Second member **38** includes a head portion **76** and a body portion **78**. A seat **80** is defined at the interface of head portion **76** and body portion **78**. Seat **80** engages a top surface **82** of hosel **28** when second member **38** is fully positioned within bore **26** of hosel **28**. As can also be seen in FIG. **3**, a bottom of club head **24** includes a threaded hole **86** which receives fastener **32** shown in FIG. **1**. This fastener **32** threadably extends through threaded hole **86** and engages a threaded hole **88** of first member **38**.

(20) As illustrated in FIGS. **1-3**, ribs **64** are axially offset from a head portion **76** of the second member **38**. A body portion **78** of the second member **38** includes a cylindrical region that is formed between the ribs **64** and the head portion **76**. As illustrated, the cylindrical region has an outer diameter that is smaller than an outer diameter of the head portion **76**.

(21) As shown in FIG. **3**, threaded hole **88** is illustrated as communicating with bore **44**. However, in other embodiments, threaded hole **88** may be a blind threaded hole which does not communicate with bore **44**. When first member **36** is fully positioned within second member **38** as shown in FIG. **1**, fastener **32** extends through threaded hole **86** and threadably engages threaded hole **88** of first member **36**. As fastener **32** is tightened, first member **36** is drawn downward along axis **30** to bring first and second axial abutment surfaces **74**, **84** into engagement. Continued tightening of fastener **32** thus biases first and second members **36**, **38** downwardly along connector axis **30** to bring seat **80** into engagement with top surface **82** of hosel **26**. The foregoing tightening of fastener **32** thus axially connects first and second members **36**, **38** and also club head **24** to club shaft **22**. As discussed above, teeth **60**, **62** engage one another to rotationally fix first connector **36** relative to second connector **38** and thus club head **24** as well. As also discussed above, ribs **64** engage club head anti-rotational feature **58** to rotationally fix second member **38** relative to club head **24**, and thus also rotationally fix first member **36** and club shaft **22** relative to club head **24**.

(22) FIGS. **4-11** show various alternative embodiments of second member which are designed to mate with the brand specific mating features of various adjustable club heads. As can readily be seen from inspection of these figures, and as introduced above, they all include a common second anti-rotational member which is the same as second anti-rotational member described above relative to FIGS. **1-3**. However, each of the embodiments of second member shown herein include a different third anti-rotational member which is specific to the specific brand of club head which it will be utilized with.

(23) Indeed, with reference to FIGS. **4-5**, a second embodiment of a second member **138** is illustrated. This second member **138** includes a head portion **176** and a body portion **178**. Second member **138** also includes second anti-rotational member **154** which includes a plurality of teeth **162** which are identical to teeth **62** described above. Additionally, second member **138** also includes a seat **180** which functions in the same manner as seat **80** described above. However, third anti-rotational member **156** in this embodiment is different than that described above. Indeed, third anti-rotational member **156** includes a plurality of downwardly extending teeth **164**. These downwardly extending teeth **164** are arranged to engage corresponding projections within an interior bore of a hosel. This engagement rotationally fixes second member **138** within such a hosel.

(24) As can be seen in FIGS. **4-5**, third anti-rotational member **156** is formed at the interface between head portion **176** and body portion **178**. Contrast this with what is shown in FIGS. **1-3**, wherein the third anti-rotational member **56** is formed at an end of body portion **76** of second member **38**. Second member **138** also includes a second axial abutment surface **184** that functions in the same manner as second axial abutment surface **84** described above.

(25) As illustrated in FIGS. **4-5**, teeth **164** have an outer diameter that is smaller than an outer diameter of the head portion **176**. Further, the outer diameter of the teeth **164** is greater than an outer diameter of the body portion **178**. Additionally, the outer diameter of the cylindrical region of body portion **178** is less than the outer diameter of the teeth **164** and head portion **176**.

(26) A third embodiment of a second connector **238** is illustrated in FIGS. **6-7**. This embodiment

also includes a head portion **276** and a body portion **278**. This embodiment also includes a second anti-rotational member **254** defined by a plurality of teeth **262** which are identical to the second anti-rotational member as described above. A seat **280** is also formed at the interface between head portion **276** and body portion **278** and functions in the same manner as the seats described above.

(27) However, in this embodiment, third anti-rotational member **256** takes the form of a plurality of facets which generally resemble the outer periphery of a nut. These facets **264** are designed to mate with facets formed within an interior bore of a club head such that rotation of second member **238** relative to club head is prevented. As can also be seen from inspection of FIGS. **6** and **7**, third anti-rotational member **256** is formed at the interface between head portion **276** and body portion **278** in a similar manner as that described above relative to FIGS. **4-5**. As clearly illustrated in FIG. **6**, the intersection of adjacent facets is stepped radially inward from the outer diameter of head portion **276** such that the facets have an outer diameter that is less than the outer diameter of head portion **276** and greater than the outer diameter of a cylindrical region of body portion **278**. As can also be seen in FIGS. **6-7**, second member **238** includes a bore **246** for receipt of first member **38** in the manner described above. Second member **238** also includes a second axial abutment surface **284** that functions in the same manner as second axial abutment surface **84** described above.

(28) Another embodiment of a second member **338** is shown in FIGS. **8-9**. This embodiment also includes a head portion **376** and a body portion **378**. A second anti-rotational member **354** is formed thereon which includes a plurality of teeth **362**. Second anti-rotational member **354** is identical to the second anti-rotational members described above.

(29) However, second member **338** includes a third anti-rotational member **356** formed directly on head portion **376**. This third anti-rotational member **356** includes a plurality of downwardly extending teeth **364**. These downwardly extending teeth **364** mate with corresponding upwardly extending teeth formed adjacent a top surface of a hosel. These teeth engage one another to prevent relative rotation of second member **338** relative to the club head. Also, second member **338** includes a bore **346** for receipt of first member **30** in the same manner as described above. Second member **338** also includes a second axial abutment surface **384** that functions in the same manner as second axial abutment surface **84** described above.

(30) FIGS. **10-11** show a fifth embodiment of a second member **438**. This embodiment includes a head portion **476** and a body portion **478**. This embodiment also includes a second anti-rotational member **454** which includes a plurality of upwardly extending teeth **462** which are identical to the second anti-rotational members described above. Second member **438** also includes a seat **480** which functions in the same manner as the seats described above.

(31) However, third anti-rotational member **456** of second member **438** is formed on body portion **478** and between head portion **476** and an end of body portion **478** along the outer periphery of body portion **478**. This embodiment of a third anti-rotational member **456** includes a plurality of outwardly extending ribs **464**. This plurality of outwardly extending ribs **464** engages a plurality of inwardly extending ribs formed along an interior surface of a hosel of a club head to thereby prevent rotation of second member **438** relative to the club head. As illustrated in FIGS. **10-11**, seat **480** of head portion **476** has an outer diameter that is greater than the outer diameter of ribs **464**. Ribs **464** have a greater outer diameter than a cylindrical region of body portion **478**. This embodiment also includes a bore **446** which receives first connector **38** in the same manner as described above. Although not illustrated, second member **438** may also include a second axial abutment surface that functions in the same manner as second axial abutment surface **84** described above.

(32) As will be understood from the foregoing, each of the embodiments shown in FIGS. **4-11** are designed to mate with a specific adjustable club head based upon the brand thereof. However, each of these embodiments will readily mate with the common first member **36** shown in FIGS. **1-3**. As a result, a user may utilize any desired shaft which incorporates first member **36** with any of the embodiments of second members described herein. As a result, a user is no longer locked into a

single brand of adjustable club heads and adjustable club shafts, but instead can readily interchange one brand of an adjustable club head with a different brand of a club shaft.

(33) Those skilled in the art will readily recognize that the installation process for these varying embodiments of second members is essentially the same as that described above relative to FIG. 3. Indeed, the second member is inserted into the hosel and arranged such that its third anti-rotational member mates with the club head anti-rotational member of the club head. The first member **38** is inserted therein. A fastener **32** is inserted through the threaded hole **86** of the club head **24** and tightened to engage the threaded hole **88** of first member **36**. Continue tightening of fastener **32** axially biases the first and second members along their connector axis until members are fully axially located. The first and second anti-rotational members engage one another to prevent relative rotation of the first member relative to the second member, and the third anti-rotational member engages the club head anti-rotational member to prevent relative rotation between these components.

(34) As a result, a user need only outfit a variety of clubs with a common style of first connector **36** as shown in FIGS. 1-3. Thereafter, and depending upon the particular style of adjustable club head they owned, they select the appropriate second connector member. Each one of these variety of shafts incorporating first member **36** will readily mate with the particular club head which receives the brand specific second member described herein.

(35) All references, including publications, patent applications, and patents cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(36) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) is to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

(37) Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. A universal connector for connecting a golf club head and a golf club shaft, the golf club head having a bore and a golf club head anti-rotation member, the connector comprising: a first member having a head portion, a body portion, a shaft bore for receiving a terminal end of the golf club shaft, and a first anti-rotational member in the form of a plurality of axially facing teeth, the head

portion having a tapered portion with an outer radial surface that reduces in outer diameter when moving axially away from the first anti-rotational member, the tapered portion being axially offset from at least a portion of the body portion, the tapered portion, body portion, and first anti-rotational member formed as a single piece, the first member having a second bore; at least one second member having a head portion, a body portion, a second anti-rotational member in the form of a plurality of axially facing teeth, a third anti-rotational member, the at least one second member being matable with the first member, the first and second anti-rotational members axially and angularly engageable with one another to prevent relative rotation of the first member relative to the second member about a longitudinal axis of the connector, the first and second anti-rotational members form a mating interface which is exposed on an exterior radially outer facing surface of the connector, the third anti-rotational member configured to mate with the golf club head anti-rotation member of the golf club head to prevent relative rotation of the second member relative to the golf club head, the third anti-rotational member axially offset from the second anti-rotational member, the head portion, body portion, second anti-rotational member, and third anti-rotational member being formed as a single piece; a mounting screw receivable in the second bore, the mounting screw configured to abut against the club head and threadedly engage the first member to axially bias the first member into the second member; wherein the third anti-rotational member of the second member is selected from the group consisting of: a plurality of radially outwardly extending ribs that are axially offset from the head portion of the second member, a cylindrical region being formed between the plurality of radially outwardly extending ribs and the head portion of the second member, the cylindrical region having an outer diameter that is smaller than an outer diameter of the head portion of the second member; a plurality of axially extending teeth that have an outer diameter that is smaller than an outer diameter of the head portion of the second member and the outer diameter of the plurality of axially extending teeth is greater than an outer diameter of the body portion of the second member; a plurality of radially outwardly facing facets, the facets being stepped radially inward from the head portion of the second member, the head portion of the second member being proximate a first end of the second member, the plurality of radially outwardly facing facets being positioned between the head portion and an opposite second end of the second member; a plurality of outwardly extending ribs formed along an outer periphery of the body portion of the second member between the head portion and an end of the body portion, the plurality of outwardly extending ribs have an outer diameter that is smaller than an outer diameter of the head portion of the second member; and a plurality of members in the form of teeth, ribs or facets that have an outer dimension that is smaller than an outer diameter of the head portion of the second member.

2. The connector of claim 1, wherein: the axially extending teeth of the first member are formed at an axial extent of the head portion proximate the body portion of the first member; the axially extending teeth of the second member are formed at an axial extent of the head portion of the second member; and the axially extending teeth of the first member are received within spaces formed between the axially extending teeth of the second member to rotationally fix the first member relative to the at least one second member.

3. The connector of claim 1, wherein the second member has a radial transition at an interface of the head portion of the second member and the body portion of the second member.

4. The connector of claim 3, wherein: the body portion of the second member extends between a free end of the second member and the interface between the head and body portions of the second member; the third anti-rotational member is located between the free end of the second member and the interface.

5. The connector of claim 1, wherein the head portion of the first member has a maximum outer diameter which is greater than a maximum outer diameter of the body portion of the first member.

6. The connector of claim 5, wherein the body portion of the first member includes a transition region and an extension region, wherein a first axial abutment surface is formed at an interface of

the transition region and the extension region.

7. The connector of claim 1, wherein the third anti-rotational member circumscribes at least a portion of the body portion of the second member.

8. The connector of claim 1, wherein the third anti-rotational member of the second member is the plurality of axially extending teeth that have the outer diameter that is smaller than the outer diameter of the head portion of the second member and the outer diameter of the plurality of axially extending teeth is greater than an outer diameter of the body portion of the second member, the plurality of axially extending teeth being formed at an interface of the head portion and the body portion of the second member.

9. The connector of claim 1, wherein the third anti-rotational member of the second member is the plurality of axially extending teeth that have the outer diameter that is smaller than the outer diameter of the head portion of the second member and the outer diameter of plurality of axially extending teeth is greater than an outer diameter of the body portion, a cylindrical region is positioned axially between the plurality of axially extending teeth and the head portion of the second member, the cylindrical region having an outer diameter that is greater than the outer diameter of the body portion and is smaller than the outer diameter of the head portion.

10. The connector of claim 1, wherein the third anti-rotation member of the second member is the plurality of radially outwardly extending ribs that are axially offset from the head portion of the second member such that the body portion of the second member forms the cylindrical region between the plurality of radially outwardly extending ribs and the head portion of the second member, the cylindrical region having the outer diameter that is smaller than the outer diameter of the head portion of the second member, the plurality of radially outwardly extending ribs are formed proximate an end of the body portion of the second member.

11. The connector of claim 1, wherein the third anti-rotation member of the second member is the plurality of radially outwardly extending ribs that are axially offset from the head portion of the second member such that the body portion of the second member forms the cylindrical region between the plurality of radially outwardly extending ribs and the head portion of the second member, the cylindrical region having the outer diameter that is smaller than the outer diameter of the head portion of the second member, and the plurality of radially outwardly extending ribs have an outer diameter that is smaller than the outer diameter of the cylindrical region.

12. The connector of claim 1, wherein the third anti-rotation member of the second member is the plurality of members in the form of teeth, ribs or facets that have the outer dimension that is smaller than the outer diameter of the head portion of the second member, and the plurality of members do not extend axially entirely to the end of the body portion of the second member.

13. The connector of claim 1, wherein the third anti-rotation member of the second member is the plurality of radially outwardly facing facets, a minimum radial dimension of the facets is smaller than the maximum radial dimension of the head portion of the second member, the facets circumscribing at least a portion of the body portion of the second member.

14. The connector of claim 1, wherein the third anti-rotation member is sized and configured to be fully received in the bore of the golf club head such that the entirety of the third anti-rotation member is fully hidden within the bore of the golf club head and is not visible.

15. A connector for connecting a golf club head and a golf club shaft, the connector comprising: a first member having a first anti-rotational member formed thereon and configured for connection to the shaft; at least one second member having a second anti-rotational member formed thereon, the at least one second member configured for mating with the golf club head, the first anti-rotational member engageable with the second anti-rotational member, the at least one second member including a head portion, a body portion and a third anti-rotational member that are formed as a single piece with the second anti-rotational member, the third anti-rotational member being selected from the group consisting of: a plurality of radially outwardly extending ribs that are axially offset from the head portion of the second member such that the body portion of the second member

forms a cylindrical region between the plurality of radially outwardly extending ribs and the head portion of the second member, the cylindrical region having an outer diameter that is smaller than an outer diameter of the head portion of the second member; a plurality of axially extending teeth that have an outer diameter that is smaller than an outer diameter of the head portion of the second member and the outer diameter of the plurality of axially extending teeth is greater than an outer diameter of the body portion of the second member; a plurality of radially outwardly facing facets formed at the interface between the head portion and the body portion of the second member, the facets circumscribing at least a portion of the body portion; a plurality of outwardly extending ribs formed along an outer periphery of the body portion of the second member between the head portion and an end of the body portion, the plurality of outwardly extending ribs have an outer diameter that is smaller than an outer diameter of the head portion of the second member; and a plurality of members in the form of teeth, ribs or facets that have an outer dimension that is smaller than an outer diameter of the head portion of the second member; wherein the first member includes a bore configured for receipt of a mounting screw extending through a portion of the golf club head and which threads into the first member, wherein tightening of the screw axially biases the first member and the at least one second member together; wherein the first anti-rotational member is a plurality of axially extending teeth formed at an axial extent of a head portion of the first member.

16. The connector of claim 15, wherein the second anti-rotational member is a plurality of axially extending teeth formed at an axial extent of a head portion of the at least one second member.

17. The connector of claim 16, wherein the axially extending teeth of the first anti-rotational member are received within spaces formed between the axially extending teeth of the second anti-rotational member to rotationally fix the first member relative to the at least one second member.

18. A universal connector for connecting a golf club head and a golf club shaft, the golf club head having a bore and a golf club head anti-rotation member, the connector comprising: a first member having a head portion, a body portion, a shaft bore for receiving a terminal end of the golf club shaft, and a first anti-rotational member in the form of a plurality of axially facing teeth, the head portion having a tapered portion with an outer radial surface that reduces in outer diameter when moving axially away from the first anti-rotational member, the tapered portion being axially offset from at least a portion of the body portion, the tapered portion, body portion, and first anti-rotational member formed as a single piece, the first member having a second bore; at least one second member having a head portion, a body portion, a second anti-rotational member in the form of a plurality of axially facing teeth, a third anti-rotational member, the at least one second member being matable with the first member, the first and second anti-rotational members axially and angularly engageable with one another to prevent relative rotation of the first member relative to the second member about a longitudinal axis of the connector, the first and second anti-rotational members form a mating interface which is exposed on an exterior radially outer facing surface of the connector, the third anti-rotational member configured to mate with the golf club head anti-rotation member of the golf club head to prevent relative rotation of the second member relative to the golf club head, the third anti-rotational member axially offset from the second anti-rotational member, the head portion, body portion, second anti-rotational member, and third anti-rotational member being formed as a single piece; a mounting screw receivable in the second bore, the mounting screw configured to abut against the club head and threadedly engage the first member to axially bias the first member into the second member; wherein the third anti-rotational member of the second member is selected from the group consisting of: a plurality of radially outwardly extending ribs that are axially offset from the head portion of the second member, a cylindrical region being formed between the plurality of radially outwardly extending ribs and the head portion of the second member, the cylindrical region having an outer diameter that is smaller than an outer diameter of the head portion of the second member; a plurality of axially extending teeth that have an outer diameter that is smaller than an outer diameter of the head portion of the second member

and the outer diameter of the plurality of axially extending teeth is greater than an outer diameter of the body portion of the second member; a plurality of radially outwardly facing facets, the facets being stepped radially inward from the head portion of the second member, the head portion of the second member being proximate a first end of the second member, the plurality of radially outwardly facing facets being positioned between the head portion and an opposite second end of the second member; a plurality of outwardly extending ribs formed along an outer periphery of the body portion of the second member between the head portion and an end of the body portion, the plurality of outwardly extending ribs have an outer diameter that is smaller than an outer diameter of the head portion of the second member; and a plurality of members in the form of teeth, ribs or facets that have an outer dimension that is smaller than an outer diameter of the head portion of the second member.
